AKARROUCH at DialectSentEval 2026: A Generate-and-Rerank Approach for Arabic Dialect Sentiment Swapping

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Abstract

We describe the AKARROUCH submission to DialectSentEval 2026 Subtask 2, which requires reversing the sentiment of an Arabic dialect sentence while preserving its content. We fine-tune AraT5v2 for direction-controlled generation and use it to produce multiple candidates. We then rerank these candidates using CAMeLBERT target-sentiment probabilities and source-to-candidate chrF. The system ranked second in the official evaluation, achieving 78.51% sentiment accuracy, 49.82 BLEU, and 70.12 chrF.

1 Introduction

DialectSentEval 2026 Subtask 2 asks systems to reverse the sentiment of an Arabic dialect sentence while retaining the information expressed by the source (Ezzini et al., 2026a). The task extends AraSentEval's work on Arabic sentiment analysis and swapping (Ezzini et al., 2026b). Our work uses a direction-controlled AraT5v2 generator followed by inference-time reranking. CAMeLBERT is used only during inference and is not jointly trained with the generator. Code and the executable notebook are available at `https://github.com/MedAkarrouch/DialectSentEval2026`.

2 Background

Sentiment swapping is a text style transfer problem in which a target attribute is changed while other content is retained as far as possible. The shared-task training and validation records contain a source sentence, its polarity, and an opposite-sentiment target, while the test records provide sources for generation. Parallel resources remain limited for Arabic. Ma'aks provides manually curated Arabic sentence pairs expressing opposite sentiments about the same subject (Mughaus et al., 2026). Prior work shows that limited parallel data can support supervised sentiment transfer (Mukherjee and Dusek, 2023).

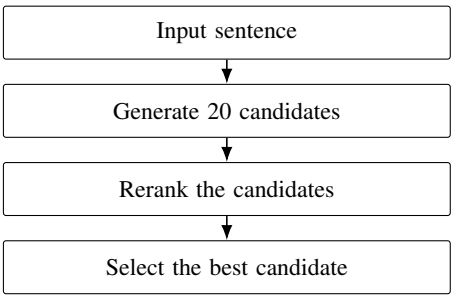


Figure 1: Overview of the generate-and-rerank inference workflow.

Generate-and-rerank methods separate candidate production from final selection. In text style transfer, Suzgun et al. (2022) generate several target-style candidates and rank them using style strength, textual similarity, and fluency. Our system follows this general structure with models and ranking signals chosen for the shared task.

3 System Description

Figure 1 gives an overview. One input produces 20 candidates, which are reranked before the highest-ranked candidate is selected.

3.1 Direction-Controlled Generation

We initialize the generator from `UBC-NLP/AraT5v2-base-1024`, an Arabic text-to-text model associated with the Octopus toolkit (Elmadany et al., 2023; UBC Deep Learning and NLP Lab, 2023). Fine-tuning treats sentiment swapping as conditional sequence generation: the source sentence and the requested direction form the model input, and the paired sentiment-swapped sentence is the target. All AraT5v2 parameters are updated.

The two transformation directions are represented by special tokens. Table 1 shows how the

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Source polarity	Target polarity	Model input
Negative	Positive	`<negative_to_positive> {source}`
Positive	Negative	`<positive_to_negative> {source}`

Table 1: Direction-controlled input templates used for fine-tuning and generation.

corresponding token is prepended to the source sentence for each direction.

The complete validation set is evaluated once per epoch for model selection. Training stops after three consecutive epochs without an improvement in validation loss, and the checkpoint with the lowest loss is used for inference. At inference, CAMeLBERT compares its positive and negative probabilities for the source. The larger value determines the inferred source polarity, so neutral does not participate in this binary decision. The opposite polarity selects the direction token.

3.2 Candidate Generation and Reranking

The selected AraT5v2 checkpoint uses deterministic beam search with 20 beams to return 20 candidates. No sampling is applied, and the method does not assume that the candidates are unique. For each candidate y_i, CAMeLBERT supplies the probability q_i of the requested positive or negative label (Inoue et al., 2021; CAMeL Lab, 2021).

The second score, h_i, is sentence-level chrF between the source and candidate (Popović, 2015). It measures character-level overlap and is used as a content-preservation proxy, not as direct evidence of semantic equivalence. During inference-time evaluation, the gold validation targets are used to calculate BLEU and chrF only after selection. They do not rank the generated candidates.

Algorithm 1 presents the complete decision procedure. Candidates are indexed in beam order. When the preceding scores are identical, the $-i$ term favors the candidate with the lower beam index. Tuple maximization is lexicographic from left to right.

The threshold separates the two ranking priorities. Candidates meeting it are ranked first by source-to-candidate chrF, then target-sentiment probability, and finally beam position. If none reaches the threshold, target-sentiment probability becomes the primary criterion, followed by chrF and beam position.

Algorithm 1 Generate-and-rerank inference

Require: source x, generator G, classifier C

$s \leftarrow \arg\max_{\ell \in \{\text{pos,neg}\}} P_C(\ell \mid x)$
$t \leftarrow \text{opposite}(s)$
$p \leftarrow \textsc{DirectionPrompt}(x, t)$
$Y \leftarrow \textsc{BeamSearch}(G, p, K = 20)$
for $i = 1, \ldots, 20$ **do**
    $q_i \leftarrow P_C(t \mid y_i)$
    $h_i \leftarrow \text{chrF}(x, y_i)$
end for
$S \leftarrow \{i : q_i \geq \tau\}$, where $\tau = 0.85$
if $S \neq \emptyset$ **then**
    $j \leftarrow \arg\max_{i \in S}(h_i, q_i, -i)$
else
    $j \leftarrow \arg\max_{1 \leq i \leq 20}(q_i, h_i, -i)$
end if
return y_j

4 Experimental Setup

We use the official training set for supervised fine-tuning and the complete validation set for epoch-level validation loss, early stopping, and checkpoint selection. Fine-tuning is performed in FP16 on two NVIDIA Tesla T4 GPUs. AdamW uses a learning rate of 5×10^{-5} and a linear schedule with 5% warmup. Appendix A reports the remaining reproducibility settings. Core software comprises PyTorch 2.10.0+cu128, Transformers 4.49.0, and SacreBLEU 2.5.1 (Post, 2018).

At inference, the selected checkpoint generates 20 candidates per input and Algorithm 1 selects one prediction. CAMeLBERT is used only for source-direction inference and candidate scoring. It contributes no AraT5v2 training loss and receives no updates. The official evaluation reports sentiment accuracy, BLEU (Papineni et al., 2002), and chrF. Sentiment accuracy evaluates the requested polarity, while BLEU and chrF measure overlap with reference outputs.

5 Results and Discussion

Table 2 presents the official test results.

Sentiment accuracy reflects how often an output received the requested polarity under the official evaluation. BLEU and chrF measure overlap with the reference outputs, with chrF providing character-level sensitivity to morphological and orthographic variation. Together, these values report the end-to-end performance of the submitted system.

Metric	Official test result
Sentiment accuracy	78.51%
BLEU	49.82
chrF	70.12

Table 2: Official DialectSentEval 2026 Subtask 2 results for Team AKARROUCH.

Transformation	Model input	Model output
Positive → Negative	هذا الموقع جدا	هذا الموقع جدا
	جميل	ممل
Negative → Positive	تطبيق متجه للفشل	تطبيق متجه للنجاح
	جدا فاشل	جدا ناجح

Table 3: Examples of sentiment swapping produced by the generate-and-rerank system.

Table 3 shows one positive-to-negative example and one negative-to-positive example. For each input, the model generates 20 candidates, reranks them using Algorithm 1, and selects the highest-ranked candidate as the output.

6 Conclusion

We presented direction-controlled AraT5v2 generation with CAMeLBERT- and chrF-based reranking. Future work could either modify fine-tuning with explicit task-specific penalty terms for sentiment reversal and content preservation, or apply preference optimization such as DPO over chosen and rejected generated candidates (Rafailov et al., 2023). Controlled inference ablations should also examine candidate count and beam width, deterministic beam search versus sampling, sentiment threshold, and length penalty.

Limitations

Because CAMeLBERT determines source polarity and scores candidates, its errors could affect both transformation-direction selection and reranking. Reliance on the same classifier used by the official sentiment metric may align the system with its decision boundary rather than broader human sentiment judgments. For the submitted model, we used a fixed inference configuration of 20 deterministic candidates, no sampling, and a 0.85 threshold. We did not conduct controlled ablations of these settings, and the selected configuration may therefore be suboptimal.

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A Implementation Details

Tables 4 and 5 list the remaining training and inference settings.

Setting	Value
Training examples	6,752
Validation examples	964
Test examples	1,931
Per-device training batch	4
Gradient accumulation	2
Weight decay	0.01
Gradient clipping	1.0
Maximum source length	256 tokens
Maximum target length	256 tokens
Maximum epochs	20
Early-stopping patience	3 epochs
Random seed	42

Table 4: Data and supervised fine-tuning configuration.

Setting	Value
Decoding method	Deterministic beam search
Number of beams	20
Returned candidates	20
Beam-search early stopping	Enabled
Length penalty	1.0
Minimum new tokens	3
Maximum new tokens	128
Target-sentiment threshold	0.85

Table 5: Inference and reranking configuration.